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MANUFACTURER'S RESPONSIBILITY

The manufacturer shall be responsible for the quality of the parts and the material supplied by him.

Since automotive brake parts are safety units and are subject to documentation, special quality assurance measures are required.

All parts must conform to the technical requirements set forth by CONTINENTAL TEVES AG in drawings and specifications (product specifications, ATE standards, DIN standards, etc.). Production supplies shall be permitted only after CONTINENTAL TEVES AG have approved the first sample.

Prior to making any change in the material (raw material, semi-finished material), its treatment and/or composition, the production method and production location, properties, identification and packing, the supplier shall advise CONTINENTAL TEVES AG, in writing, early enough to allow a new sampling to take place, if necessary. The obligation to notify CONTINENTAL TEVES AG shall also apply to changes, the effects of which are still within the agreed tolerance range (e.g. significant changes of variability bandwidth, variations of the mean value).

A change notification shall not authorize the supply of already changed parts.

In confirming the order, the manufacturer undertakes to maintain the technical delivery specifications mentioned in the above paragraph.

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2 DEFINITIONS

ABS Anti-Blockiersystem
BLS Brake Light Switch
BMR Bremsmomentenregelung (Entspricht EDS)
CS Checksum
DTC Diagnostic Trouble Code
EBD Electronic Brakeforce Distribution (Deutsch EBV)
EBV Elektronsche Bremskraftverstärkung (Engl. EBD)
EDS Elektronische Differential-Sperre (= BTCS = BASR)
ECU Electronic controller unit (ABS-Controller)
EEPROM Electrical Erasable Read-Only-Memory
IFL Inlet valve front left
IFR Inlet valve front right
IRL Inlet valve rear left
IRR Inlet valve rear right
LPFV Low Pressure Feed Valve
OFL Outlet valve front left
OFR Outlet valve front right
ORL Outlet valve rear left
ORR Outlet valve rear right
PID Parameter-Identifier
RAM Random Access Memory (volatile)
RC Response Code
SDL Serial Data Link
SID Service Identification
SN Service Name

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3 CHANGE HISTORY

12.05.2005
- Do not support security access
- Do not support any common ID-handling (coding-data etc.)
- No speed limitation in diagnostic session 87 (“DIA-Mode”)
- New local identifier
- Rearrange chapters 7.10 and 7.11
- DTC-codes beginning with 0x40xx
- DTC-table new

13.05.2005
- Delete command to WRITE PRODUCTION DATA

17.05.2005
- delete LTCS/TCS and ESP
- abbreviations EBD and BMR explained

18.05.2005
- MCI-Valve 2 and LPFV2 for Input-/Output-control

13.06.2005
Chapter 7.8.1 MCI_1-Valve and LPF_2 Valve are also needed for bleed-process

22.06.2005
Chapter 5.3 Introduce Parameter Transfer-Mode
Chapter 7.1 Introduce Parameter Transfer Mode
Chapter 7.2 Introduce Parameter Transfer Mode
Rename Chapter
Chapter 7.9.4 added (Get Parameter Block-Addresses)
Chapter 7.12.2 introduce Singel-Byte-Write Command

28.06.2005
Chapter 7.9.4 3 addressbytes each for start- and stop address

04.08.2005
Chapter 6.1 DTC 40FE added (EOL-feature)
Chapter 7.11.4 added

05.08.2005
Chapter 6.1 DTC 40FD added (Can_bus_off)

08.08.2005
Chapter 7.11.1 without any security access
Chapter 7.11.2 only to be written in extended session

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15.08.2005
Chapter 7.9.1 Add EDS-switch
Chapter 7.4 EDS – status and switch introduced

08.09.2005
Chapter 6.1 DTC 4059 as Parameter Error added

Revision 1.2

11.11.2005
Chapter 6.1 DTC 40FC as EDS-switch Error added

Wrong revision number in footer !!!!!!!!!!! >>>

Revision 1.3 = Revision 1.2

Wrong revision number in footer of previous revision !!!!!!!!!!! >>>

Revision 1.4

30.11.2005
Chapter 7.4 Operating Mode Low Byte :
RTA-bit always set (RTA is available)

Revision 1.5

23.07.2006 remove "DRAFT" from header

Revision 1.6

01.08.2006
Chapter 4.1.4 remove response 23 (routine not complete)
Chapter 7.2 description of neg. responses changed

Revision 1.7

Chapter 6.1 DTC 40FB "Disc Temperature High" added

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4 GLOBAL DIAGNOSTIC REQUIREMENTS

4.1 Conditions for Serial Communication

After Ignition-ON the ECU will wait 100ms and then start monitoring the K-line for commands with the specified data-format. If a command with the correct format is received the ECU will always respond.

4.1.1 Communication Structure

The diagnostic-tester transmits messages to the ECU and receives the response messages from the ECU.
The diagnostic tester is client, the ECU is server in SDL-Mode.

4.1.2 Message structure

Every message contains a variable number of bytes. The maximum number of bytes for the controller is n=32.

Byte	1	Format Byte	constant value : $80
	2	Target Address	ECU – Address = $29 / Tester –Address = $F0 … $FD
	3	Source Address	ECU – Address = $29 / Tester –Address = $F0 … $FD
	4	Length-Byte	Amount of data-bytes following excluding the checksum-byte
	5	Service Name	Service- /Service-response-ID
	6		Data1
...			
	n-1		Data
	n	Checksum byte	EXOR-op. of all bytes in the message excluding checksum

In this document the individual service description below will start with “Header” (containing bytes 1- 3) followed by the service-name and the databytes with checksum-byte at the end.

*Tester Request

Position	Contents	Value	Description
1	Header	0x80	Format-Byte
2		0x29	Target Address (ECU)
3		0xF?	Source Address (Tester)

*ECU Response

Position	Contents	Value	Description
1	Header	0x80	Format-Byte
2		0xF?	Target Address (Tester)
3		0x29	Source Address (ECU)

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4.1.3 Byte Format and Timing

Byteformat: 1 Startbit (logic 0),
8 Databit, (value 0 = logic 0 / value 1 = logic 1)
1 Parity-Bit (even)
1 Stopbit (logic 1)

Baudrate : 9600 Baud +/- 2%

Timing: Timeout = 5s (+/- 1s)

P1 : Interbyte gap (ECU Response) :
min. 0 ms max. 20 ms
P2 : Intermessage gap (Tester-Request - ECU-Response) :
min. 25 ms max. 199 ms
P3 Intermessage gap (ECU-Response - Tester-Request)* :
min. 31 ms max. 5 s
P4 : Interbyte gap (Tester-Request)* :
min. 0 ms max. 20 ms

*
Has to be guaranteed by the tester.
The tester has to take care that the bus is idle for at least 31 ms before addressing the TEVES -ECU.

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4.1.4 General Response - Message and Response Codes

General Response

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of Bytes following
5	Service Name	0x7F	General Response
6	Databyte 1	ID_IN	Service ID (which is rejected)
7	Databyte 2	RC	Response Code
8	Checksum	CS	EXOR

Response Codes :

$10 GENERAL REJECT
The Request received is rejected by the ECU without specifying the reason therefore.

$11 UNKNOWN COMMAND / SERVICE NOT SUPPORTED
The ECU does not support the received message (Service)

$12 SUB-FUNCTION NOT SUPPORTED / INVALID DATA OR INVALID COMMAND
The ECU supports the received message , but arguments contained within the message are not supported, or are not in the proper format

$21 BUSY , REPEAT REQUEST
The ECU is currently busy and is unable to service the request at this time.

$22 CONDITION NOT CORRECT OR SEQUENCE ERROR
The ECU could not service the request, because one or more prerequisite conditions required for its support have not been satisfied.

$31 REQUEST OUT OF RANGE
Failure-code does not exist

$33 SECURITY ACCESS DENIED
The security access level of the ECU does not allow the requested Procedure (not supported)

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$78 COMMAND ACCEPTED, RESPONSE PENDING
That happens if the response time exceeds the specified value.
e.g.
- Another EEPROM-Access has to be finished before, the request can be proceeded.
- Due to the amount of bytes to be handled, the access time is longer than the specified response-time (4 bytes can be read/written within 10 ms).

Note :
First the ECU checks whether the mode is correct to accept the incoming message. In case the ECU-mode is not correct, the request will be rejected with RC = $22. If the service-name is correct, but if the byte sequence which follows the service-name-byte (6th Byte in every tester command) is not valid, the ECU will always reject with RC = $12.

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5 ECU STATES / MODES

5.1 Operational Mode

After Power On the ECU is in Normal-Operation Mode and ready for serial communication. If the conditions (see chapter 4.1) are fulfilled Diagnostic Mode can be called up from here.

If there is any control-cycle active the ECU will reject every incoming command with the General Response "BUSY" (RC = $21).

Exception : The tester present message will always be responded.

5.2 Diagnostic Mode

5.2.1 Conditions to enter Diagnostic Mode

- The ECU detects no ABS/EDS-event.
- There is no test cycle in progress.
- The wheel velocities do not exceed 10 kph.
- The START-DIAGNOSTIC-Session Command was received correctly.

After accepting the Diagnostic-Entry-Command the ECU is in Diagnostic Mode.
There is no ABS/EDS - control-cycle possible.
Failures occurring in Diagnostic-mode are not stored to EEPROM. They are only written to RAM (volatile store) which is cleared after system-reset or power-off.

5.3 Parameter Transfer Mode

5.3.1 Conditions to enter Parameter Transfer Mode

- The ECU is in Diagnostic Mode
- The wheel velocities are 0 kph.
- The START-PARAMETER TRANFER-Session Command was received correctly.

If the ECU is in PARAMETER-TRANSFER Mode it is possible to write modified control-parameters to EEPROM via Diagnostic Tester (also see chapter 7.12.2).
Prior to entering PARAMETER-TRANSFER mode, the tester has request EEPROM-start and –stop –address from the ECU (see chapter 7.9.4)
In this mode there are only 3 commands which are accepted by the controller :

- Read EEPROM-Data by Address (chapter 7.12.1).
- Write EEPROM-Data by Address (chapter 7.12.2).
- Return to normal operation mode (chapter 7.2)

5.4 Conditions to return to Normal Mode

- No valid tester - command has been received for 5 seconds.
- A command to return to normal operation has been received. (see chapter 7.2).

Leaving Diagnostic Mode means for the controller to make a software-reset.

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5.5 Diagnostic Security Access Modes not supported

This mode is intended to unlock the ECU in order to execute special commands which are security-relevant.

Therefore the ECU has to be requested to send a seed code (Step 1).
After receiving the seed the tester has to calculate a key code and send this key to the ECU (Step2). The ECU compares the received keycode with is internally calculated one. If this comparison is successful (client and server have calculated the same value based on the seed) then the Security Access Mode will be granted.

The ECU provides 2 Access Modes (Levels) :
LEVEL B (low priority) : Level for manufacturing purposes
LEVEL D (high priority) : Provided for development purposes - all services available

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6 FAILURE HANDLING

6.1 List of Diagnostic Trouble Codes (DTCs)

0x4059	Configuration Error
Wheel 1	
0x4090	Sensor el. Fault front left
0x4091	Sensor Extrapolation Fault front left
0x4092	Sensor Periodic Fault front left
0x4093	Sensor Start Recognition Fault front left
0x4094	Sensor Phase-Length-Supervision Fault front left
0x4095	Sensor Double Frequency Check
Wheel 2	
0x40A0	Sensor el. Fault front right
0x40A1	Sensor Extrapolation Fault front right
0x40A2	Sensor Periodic Fault front right
0x40A3	Sensor Start Recognition Fault front right
0x40A4	Sensor Phase-Length-Supervision Fault front right
0x40A5	Sensor Double Frequency Check
Wheel 4 (hl)	
0x40B0	Sensor el. Fault rear left
0x40B1	Sensor Extrapolation Fault rear left
0x40B2	Sensor Periodic Fault rear left
0x40B3	Sensor Start Recognition Fault rear left
0x40B4	Sensor Phase-Length-Supervision Fault rear left
0x40B5	Sensor Double Frequency Check
Wheel 3	
0x40C0	Sensor el. Fault rear right
0x40C1	Sensor Extrapolation Fault rear right
0x40C2	Sensor Periodic Fault rear right
0x40C3	Sensor Start Recognition Fault rear right
0x40C4	Sensor Phase-Length-Supervision rear right
0x40C5	Sensor Double Frequency Check
0x40F0	Pump defective
0x40F1	Pump Connection
0x40F2	Hardware Fault
0x40F4	Voltage low
0x40F5	Internal Fault
0x40F7	Voltage high
0x40FB	Disc Temperature High
0x40FC	EDS-Switch-Failure
0x40FD	Can_bus_off
0x40FE	EOL-Test not executed
0x40FF	Speed Limit exceeded

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7 DIAGNOSTIC SERVICES

7.1 Start Diagnostic Session

Supported in Mode: 0x81, 0x87

Tester - Request

Position	Content	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x10	Service ID: Start Diagnostic Session
6	Databyte 1	0x??	Supported Diagnostic Modes (Sessions): 0x81, 0x87
7	Checksum	CS	

ECU – Modes:

0x81 Default/Standard Mode (Normal Operation Mode)
Starts 2 seconds after IGN ON
ECU control arbitration is active

0x85 Parameter-Transfer-Mode
No ECU control arbitration – no control function available
Only available when invoking from mode 87 when vehicle is not driving.

0x87 ECU Adjustment Mode
No ECU control arbitration – no control function available

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x50	Service Response ID: Start Diag.Session
6	Databyte 1	0x??	Supported Diagnostic Modes (Sessions): 0x81, 0x85, 0x87
7	Checksum	CS	

Note : Request to enter session 81 (Sequence 0x10, 0x81) will always be responded affirmatively (if no control-cycle active) without any ECU – reaction.

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Negative Response

Current ECU-State / Mode	**Valid ECU-RC**	**Reason**
Operational Mode (0x81)	0x21 0x22	ECU is executing a control-cycle Tester is requesting mode 0x87: -wheel velocities exceeding 10kph - mode 85 is requested
Parameter-Transfer-Mode (0x85)	0x22	- mode 81/mode 87 is requested
Diagnostic Mode (0x87)	0x22	- mode 85 is requested and vehicle not in standstill

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7.2 Return to Normal Operation Mode

Supported in Mode: 0x85, 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x01	Amount of valid bytes following
5	Service Name	0x20	Service ID: Stop Diagnostic Session
6	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x01	Amount of valid bytes following
5	Service Name	0x60	Service Response ID: Stop Diag. Session
6	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	0x21 0x22	ECU is executing a control-cycle Conditions not correct
Parameter-Transfer-Mode (0x85)	0x21	ECU has to finish a current procedure (EEPROM-access)
Diagnostic Mode (0x87)	0x21	ECU has to finish a current procedure (EEPROM-access)

Having accepted this command the ECU will start control-arbitration again after approximately 2 seconds.

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7.3 Tester Present Message

Supported in Mode: 0x81, 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x01	Amount of valid bytes following
5	Service Name	0x3E	Service ID: Tester Present
6	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x01	Amount of valid bytes following
5	Service Name	0x7E	Service Response ID: Tester Present
6	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)		
Diagnostic Mode (0x87)		

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7.4 Read Status of Diagnostic Trouble Code

Supported in Mode: 0x81, 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x17	Service ID: Read Status of DTC
6	Databyte1	0x??	DTC High Byte
7	Databyte2	0x??	DTC Low Byte
8	Checksum	CS	

Valid Range of DTC: 0x4090 – 0x40FF

ECU – Response if conditions are correct and requested DTC is not stored

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x57	Service Response ID: Read Status of DTC
6	Databyte1	0x00	Number Of DTC
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x0D	Amount of valid bytes following
5	Service Name	0x57	Service Response ID: Read Status of DTC
6	Databyte1	0x01	Number Of DTC
7	Databyte2	0x??	DTC High Byte
8	Databyte3	0x??	DTC Low Byte
9	Databyte4	0x??	DTC Status Byte
10	Databyte5	0x??	Fault Occurrence Counter
11	Databyte6	0x??	Fault Ignition Counter (Init Value is 40)
12	Databyte7	0x00	reserved
13	Databyte8	0x00	reserved
14	Databyte9	0x??	Vehicle velocity (1 LSB = 1km/h)
15	Databyte10	0x??	Status-Byte (Switches)
16	Databyte11	0x??	Operating Mode High Byte
17	Databyte12	0x??	Operating Mode Low Byte
18	Checksum	CS	

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Negative Response

Current ECU-State / Mode	**Valid ECU-RC**	**Reason**
Operational Mode (0x81)	0x12 0x21	DTC is not valid / not stored ECU is executing a control cycle
Diagnostic Mode (0x87)	0x12 0x21	DTC is not valid / not stored Another EEPROM Access is active

DTC-Status-Byte
Bits 3 - 0: not

Bit 4: Readiness-Flag / not supported

Bits 6,5: Storage State
- 00 Failures has not occurred up to now
- 01 No current fault, but fault stored in EEPROM
- 10 Failure detected (RAM), but not stored in EEPROM yet
  This mode is not supported by the TEVES ECU.*)
- 11 Fault stored in RAM (current fault) and in EEPROM

Bit 7: Warning-Lamp Calibration Status not supported

*) Even for failures occurring during diagnostic-mode the bits 5 and 6 will show "1" (although failures occurring during DIA-Mode are stored to RAM only!!).

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SWITCH - Status-Byte
Bits are set to "1", if

Bit 0: Brake-Light - Switch activated
Bit 1: not used
Bit 2: not used
Bit 3: EDS Switch output *
Bit 4: not used
Bit 5: not used
Bit 6: not used
Bit 7: not used
*if EDS is configured

Operating Mode :

High Byte:
Bit 0: ABS control active
Bit 1: EDS active *
Bit 2: not used
Bit 3: not used
Bit 4: not used
Bit 5: not used
Bit 6: not used
Bit 7: not used
*if EDS is configured

Low Byte:
Bit 0: not used
Bit 1: not used
Bit 2: RTA active (set to 1, RTA always available)
Bit 3: First Run (Run-up - Mode)
Bit 4: not used
Bit 5: EBD control active
Bit 6: not used
Bit 7: not used

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7.5 Read Diagnostic Trouble Codes (EEPROM and RAM)

Supported in Mode: 0x81, 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		Description
4	Length Byte	0x04	Amount of valid bytes following
5	ServiceName	0x18	Service ID: Read DTCs by Status
6	Databyte1	0x00 0x01 0x02	>> Positive response with no DTCs stored >> regular request for 2byte Hex-Coded DTCs
7	Databyte2	0xFF	DTC Group High Byte
8	Databyte3	0xFF	DTC Group Low Byte
9	Checksum	CS	

ECU - Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	N*3+2	Amount of valid bytes following
5	ServiceName	0x58	Service Response ID: Read DTCs by Stat.
6	Databyte1	N	Amount of stored DTCs
7	Databyte2	DTC1	DTC1 High Byte
8	Databyte3	DTC1	DTC1 Low Byte
9	Databyte4	Status	Status Byte of DTC1 – see Chapter above
10	Databyte5	DTC2	DTC2 High Byte
11	Databyte6	DTC2	DTC2 Low Byte
12	Databyte7	Status	Status Byte of DTC2 – see Chapter above
13	Databyte8	DTC3	DTC3 High Byte
…	…	…	…
3*N+4	Databyte (3*N-1)	DTC_N	DTC_N High Byte
3*N+5	Databyte (3*N)	DTC_N	DTC_N Low Byte
3*N+6	Databyte (3*N+1)	Status	Status Byte of DTC_N – see above
3*N+7	Checksum	CS	

DTC1 is oldest DTC DTC_N is latest one.

The ECU provides a fifo-buffer for 10 DTCs.

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Negative Response

Current ECU-State / Mode	**Valid ECU-RC**	**Reason**
Operational Mode (0x81)	0x12 0x21	DTC-Group not supported ECU is executing a control cycle
Diagnostic Mode (0x87)	0x12 0x21	DTC-Group not supported Another EEPROM Access is active

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7.6 Clear all Diagnostic Trouble Codes (EEPROM and RAM)

Supported in Mode: 0x81, 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x14	Service ID: Clear stored DTCs
6	Databyte1	0xFF	DTC Group High Byte *
7	Databyte2	0xFF	DTC Group Low Byte *
8	Checksum	CS	

Group 0xFFFF = all DTCs

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x54	Service ID: Clear stored DTCs
6	Databyte1	0xFF	Group of DTC High Byte
7	Databyte2	0xFF	Group of DTC Low Byte
8	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	0x12 0x21	DTC-Group not supported ECU is executing a control cycle
Diagnostic Mode (0x87)	0x12 0x21	DTC-Group not supported Another EEPROM Access is active

Receiving the CLEAR-command during operational mode ($81) all historical DTCs will be cleared. DTCs which have been detected during the current ignition cycle will not be affectetd.

Clearing during diagnostic mode ($87) will delete all current and historical DTCs.

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7.7 Security Access – Procedure / not supported

Supported in Mode: 0x87

7.7.1 Step 1: Request Seed Code from ECU

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x27	Service ID: Request Security Access
6	Databyte1	(D1)	Access Level
7	Checksum	CS	

Access Level Byte D1 (Seed-Request):
$03 Security Access Level B
$07 Security Access Level D

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x04	Amount of valid bytes following
5	Service Name	0x67	Service Response ID: Security Access
6	Databyte1	(D1)	Access Level
7	Databyte2	(SB1)	Seed High Byte
8	Databyte3	(SB2)	Seed Low Byte
9	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	0x21 0x22	ECU is executing a control-cycle Conditions not correct
Diagnostic Mode (0x87)	0x12	Wrong access level

If the ECU already has the desired access level it will return 0x0000 as seed code.

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7.7.2 Step 2: Send Calculated Key Code to ECU / not supported

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x04	Amount of valid bytes following
5	Service Name	0x27	Service ID: Request Security Access
6	Databyte1	(D2)	Access Level
7	Databyte2	(KB1)	Key Code High Byte
8	Databyte3	(KB2)	Key Code Low Byte
9	Checksum	CS	

Access-Level-Byte D2 (Send-Key):

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x67	Service Response ID: Security Access granted
6	Databyte1	(D2)	Access Level
7	Databyte2	0x34	Access allowed
8	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	0x21 0x22	ECU is executing a control-cycle Conditions not correct
Diagnostic Mode (0x87)	0x22 0x12 0x10	No seed requested (Step 1) Wrong access level Invalid Key

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7.8 Input- / Output Control by Local Identifiers (LIDs)

Supported in Mode: 0x87

7.8.1 Valves- and Pump- Control (Output Actuations)

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x08	Amount of valid bytes following
5	Service Name	0x30	Service ID: I/O-Control by Local ID
6	Databyte1	0x01	ID Output Actuation
7	Databyte2	0x07	Control-Option: Short Term Adjustment
8	Databyte3	0x??	Pattern A: Control Byte 1
9	Databyte4	0x??	Pattern A: Control-Byte 2
10	Databyte5	0x??	Timer Value for Pattern A / 1 LSB = 10ms
11	Databyte6	0x??	Pattern B: Control Byte 1
12	Databyte7	0x??	Pattern B: Control-Byte 2
13	Checksum	CS	

Contents of Control Bytes

Byte 1:
- Bit 0 Inlet Valve Front Left (IFL)
- Bit 1 Outlet Valve Front Left (OFL)
- Bit 2 Inlet Valve Front Right (IFR)
- Bit 3 Outlet Valve Front Right (OFR)
- Bit 4 Inlet Valve Rear Left (IRL)
- Bit 5 Outlet Valve Rear Left (ORL)
- Bit 6 Inlet Valve Rear Right (IRR)
- Bit 7 Outlet Valve Rear Right (ORR)

Byte 2:
- Bit 0 Master Cylinder Isolation Valve 1 (MCI V1)
- Bit 1 Master Cylinder Isolation Valve 2 (MCI V2)
- Bit 2 Low Pressure Feed Valve 1 (LPF V1)
- Bit 3 Low Pressure Feed Valve 2 (LPF V2)
- Bit 4 Pump
- Bit 5 reserved
- Bit 6 reserved
- Bit 7 reserved

For output activation the specified bits have to be set to "1".

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The outputs will be activated with the pattern A-value as long as defined via timer –value (Data Byte 7). If the timer expires, the outputs will be activated to pattern B for 2 seconds unless there is another activation-command.
The tester-software has to make sure, that no output is activated for more than 90 seconds.
See TEVES document ATE N 553 93.64.

Timer Value 0 will deactivate pattern A so that after receiving the command the ECU will control the outputs according to pattern B for 2 seconds.

ECU – Response if conditions are correct

Position	**Contents**	**Value**	**Description**
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x70	Service Response ID: I/O-Ctrl.by Local ID
6	Databyte1	0x01	ID Output Actuation
7	Databyte2	0x07	Control-Option: Short Term Adjustment
8	Checksum	CS	

Negative Response

Current ECU-State / Mode	**Valid ECU-RC**	**Reason**
Operational Mode (0x81)	$21 $22	ECU is executing a control cycle Conditions not correct
Diagnostic Mode (0x87)	$21 $22	Busy with pattern A – try again when timer has expired Conditions not correct (e.g. system switch off / run-up-mode)

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7.9 Read Internal Data (Status) By Local Identifier (LIDs)

Supported in Mode: 0x81, 0x87

7.9.1 Switch Status

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x21	Service ID: Read Data by Local ID
6	Databyte1	0x01	ID: Control Switch Status
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x61	Service Resp. ID: Read Data by Local ID
6	Databyte1	0x01	ID: Control Switch Status
7	Databyte2	0x??	Switch Status Byte
7	Checksum	CS	

Switch Status Byte
Bit 0 Brake Light Switch
Bit 1 0 / reserved
Bit 2 EDS Switch*
Bit 3 EDS functionality available*
Bit 4 0 / reserved
Bit 5 0 / reserved
Bit 6 0 / reserved
Bit 7 0 / reserved

* if EDS is configured

Bits are set to "1" if switches are activated

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	ECU is executing a control cycle
Diagnostic Mode (0x87)		

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7.9.2 Read Supply Voltages

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x21	Service ID: Read Data by Local ID
6	Databyte1	0x02	ID: Sensor-Inputs and Voltages
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x05	Amount of valid bytes following
5	Service Name	0x61	Service Resp. ID: Read Data by Local ID
6	Databyte1	0x02	ID: Sensor-Inputs and Voltages
7	Databyte2	0x??	Reference Voltage (V_ref) 1 LSB = 39 mV
8	Databyte3	0x??	Pump 1 LSB = 57,7 mV
9	Databyte4	0x??	Connector PIN 30 1 LSB = 70,8 mV
10	Checksum	CS	

If sensor-values of configured sensors are not available/reliable the ECU will read out 0xFFFF

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	ECU is executing a control cycle
Diagnostic Mode (0x87)		

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7.9.3 Wheel Speed Sensor Inputs

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x21	Service ID: Read Data by Local ID
6	Databyte1	0x03	ID: Wheel Sensor-Inputs
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x0A	Amount of valid bytes following
5	Service Name	0x61	Service Resp. ID: Read Data by Local ID
6	Databyte1	0x07	ID: Wheel Sensor-Inputs
7	Databyte2	0x??	Front Left Sensor (High Byte)
8	Databyte3	0x??	Front Left Sensor (L-Byte) 1LSB= 0.01kph
9	Databyte4	0x??	Front Right Sensor (High Byte)
10	Databyte5	0x??	Front Right Sens. (L-Byte) 1LSB= 0.01kph
11	Databyte6	0x??	Rear Left Sensor (High Byte)
12	Databyte7	0x??	Rear Left Sensor (L-Byte) 1LSB= 0.01 kph
13	Databyte8	0x??	Rear Right Sensor (High Byte)
14	Databyte9	0x??	Rear Right Sens. (L-Byte) 1LSB= 0.01kph
15	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	ECU is executing a control cycle
Diagnostic Mode (0x87)		

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7.9.4 Get Parameter-Block-Addresses (EEPROM Start-address/End-address)

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x21	Service ID: Read Data by Local ID
6	Databyte1	0x0F	ID: Parameterblock-addresses
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x08	Amount of valid bytes following
5	Service Name	0x61	Service Resp. ID: Read Data by Local ID
6	Databyte1	0x0F	ID: Parameterblock-addresses
7	Databyte2	0x??	EEPROM-Start-address (High Byte)
	Databyte3	0x??	EEPROM-Start-address
8	Databyte4	0x??	EEPROM-Start-address (Low-Byte)
9	Databyte5	0x??	EEPROM-End-address (High Byte)
	Databyte6	0x??	EEPROM-End-address
10	Databyte7	0x??	EEPROM-End-address (Low-Byte)
11	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	ECU is executing a control cycle
Diagnostic Mode (0x87)		

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7.10 Read ECU Identification Data Services

Supported in Mode: 0x81, 0x87

7.10.1 Read ECU Identification Data Table

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x1A	Service ID: Read ECU Identification
6	Databyte1	0x80	ID: ECU-Identification Data Table
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x20	Amount of valid bytes following
5	Service Name	0x5A	Service Response ID: Read ECU Identification
6	Databyte1	0x80	ID: ECU-Identification Data Table
7	Databyte2	0x??	ECU Ident. Data Byte 1
8	Databyte3	0x??	ECU Ident. Data Byte 2
…	Databyte..	…	…
36	Databyte30	0x??	ECU Ident. Data Byte 30
37	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	- ECU is executing a control cycle - ECU is proceeding another EEPROM-access
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.10.2 Read ContiTEVES Production Data

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x1A	Service ID: Read ECU Identification
6	Databyte1	0x8A	ID: Production Data
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x20	Amount of valid bytes following
5	Service Name	0x5A	Service Response ID: Read ECU Identification
6	Databyte1	0x8A	ID: Production Data
7	Databyte2	0x??	Production Data Byte 1
8	Databyte3	0x??	Production Data Byte 2
…	Databyte..	…	…
36	Databyte31	0x??	Production Data Byte 30
37	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	- ECU is executing a control cycle - ECU is proceeding another EEPROM-access
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.10.3 Read Supplier-ECU-Software-ID

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x1A	Service ID: Read ECU Identification
6	Databyte1	0x8B	ID: Supplier ECU-Software ID
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x20	Amount of valid bytes following
5	Service Name	0x5A	Service Response ID: Read ECU Identification
6	Databyte1	0x8B	ID: Supplier ECU Software ID
7	Databyte2	0x??	Supplier-ECU-ID-Byte 1
8	Databyte3	0x??	Supplier-ECU-ID-Byte 2
…	Databyte..	…	…
36	Databyte31	0x??	Supplier-ECU-ID-Byte 30
37	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	- ECU is executing a control cycle
Diagnostic Mode (0x87)		

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ID-Data-Byte	Content
01	No.of version-name bytes (ASCII) following (01hex - 0Chex)
02	version-name.byte1
03	version-name.byte2
04	version-name.byte3
05	version-name.byte4
06	version-name.byte5
07	version-name.byte6
08	version-name.byte7
09	version-name.byte8
10	version-name.byte9
11	version-name.byte10
12	version-name.byte11
13	version-name.byte12
14	Address of parameter-adjustment system (High - Byte)
15	Address of parameter-adjustment system (Low - Byte)
16	Amount of production-data following (0Ehex)
17	reserved
18	reserved
19	reserved
20	reserved
21	reserved
22	reserved
23	Additional Index (Software-Version-Number)
24	reserved
25	reserved
26	reserved
27	reserved
28	reserved
29	reserved
30	reserved

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7.10.4 Read Vehicle Information (VIN) Data

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x1A	Service ID: Read ECU Identification
6	Databyte1	0x90	ID: Vehicle Information (VIN) Data
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x1B	Amount of valid bytes following
5	Service Name	0x5A	Service Response ID: Read ECU Identification
6	Databyte1	0x90	ID: Vehicle Information (VIN) Data
7	Databyte2	0x??	VIN Data Byte 1
8	Databyte3	0x??	VIN Data Byte 2
…	Databyte..	…	…
31	Databyte26	0x??	VIN Data Byte 25
32	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	- ECU is executing a control cycle - ECU is proceeding another EEPROM-access
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.10.5 Read Programming Date

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x1A	Service ID: Read ECU Identification
6	Databyte1	0x99	ID: Programming Date
7	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x06	Amount of valid bytes following
5	Service Name	0x5A	Service Response ID: Read ECU Identification
6	Databyte1	0x99	ID: Programming Date
7	Databyte2	0x??	Programming Date Byte 1 (Century)
8	Databyte3	0x??	Programming Date Byte 2 (Year)
9	Databyte4	0x??	Programming Date Byte 3 (Month)
10	Databyte5	0x??	Programming Date Byte 4 (Day)
11	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	- ECU is executing a control cycle - ECU is proceeding another EEPROM-access
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.11 Write Data by Local Identifiers

7.11.1 Write ECU Identification Data Table

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x20	Amount of valid bytes following
5	Service Name	0x3B	Service ID: Write Data by Local Identifier
6	Databyte1	0x80	ID: ECU-Identification Data Table
7	Databyte2	0x??	ECU Ident. Data Byte 1
8	Databyte3	0x??	ECU Ident. Data Byte 2
…	Databyte..	…	…
34	Databyte30	0x??	ECU Ident. Data Byte 30
35	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x7B	Service Response ID: Write Data by Local Identifier
6	Databyte1	0x80	ID: ECU-Identification Data Table
7	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21 $22	- ECU is executing a control cycle - Conditions not correct
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.11.2 Write Vehicle Information (VIN) Data

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x1B	Amount of valid bytes following
5	Service Name	0x3B	Service ID: Write Data by Local Identifier
6	Databyte1	0x90	ID: VIN Data
7	Databyte2	0x??	VIN Data Byte 1
8	Databyte3	0x??	VIN Data Byte 2
…	Databyte..	…	…
31	Databyte26	0x??	VIN Data Byte 25
32	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x7B	Service Response ID: Write Data by Local Identifier
6	Databyte1	0x90	ID: VIN Data
7	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21	- ECU is executing a control cycle
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.11.3 Write Programming Date

Supported in Mode: 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x06	Amount of valid bytes following
5	Service Name	0x3B	Service ID: Write Data by Local Identifier
6	Databyte1	0x99	ID: Programming Date
7	Databyte2	0x??	Programming Date Byte 1 (Century)
8	Databyte3	0x??	Programming Date Byte 2 (Year)
9	Databyte4	0x??	Programming Date Byte 3 (Month)
10	Databyte5	0x??	Programming Date Byte 4 (Day)
11	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x7B	Service Response ID: Write Data by Local Identifier
6	Databyte1	0x99	ID: Programming Date
7	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21 $22	- ECU is executing a control cycle - Conditions not correct
Diagnostic Mode (0x87)	$21	- ECU is proceeding another EEPROM-access

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7.11.4 Confirm EOL-Activities (Reset EOL-Flag)

Supported in Mode: 0x87

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x03	Amount of valid bytes following
5	Service Name	0x3B	Service ID: Write Data by Local Identifier
6	Databyte1	0xFE	ID: EOL-Procedure
7	Databyte2	0x00	EOL-Confirmation (EOL tests done)
11	Checksum	CS	

ECU – Response if conditions are correct

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x02	Amount of valid bytes following
5	Service Name	0x7B	Service Response ID: Write Data by Local Identifier
6	Databyte1	0xFE	ID: EOL-Procedure
7	Checksum	CS	

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21 $22	- ECU is executing a control cycle - Conditions not correct
Diagnostic Mode (0x87)	$21 $31	- ECU is proceeding another EEPROM-access - Databyte 2 has wrong value

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7.12 ECU – Services for Development

Supported in Mode: 0x81, 0x85, 0x87

7.12.1 Read EEPROM Data

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x06	Amount of valid bytes following
5	Service Name	0x23	Service ID: Read Memory by Address
6	Databyte1	0x0?	Address High Byte
7	Databyte2	0x??	Address Mid. Byte
8	Databyte3	0x??	Address Low Byte*
9	Databyte4	0x03	EEPROM Access
10	Databyte5	N	Number of Bytes to be read out Range 0x01 – 0x04
11	Checksum	CS	

ECU – Response if conditions are correct:

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	N+1	Amount of valid bytes following
5	Service Name	0x63	Service Response ID: Read Memory by Address
6	Databyte1	0x??	EEPROM Value 1
…	…	…	…
5+N	DatabyteN	0x??	EEPROM Value N
6+N	Checksum	CS	

*1 KB-EEPROM : Address-range 0x000 – 0x3FF

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21 $31	- ECU is executing a control cycle - ECU is executing an EEPROM-Access - Wrong address range amount of bytes not correct
Diagnostic Mode (0x87)	$21 $31	- ECU is executing an EEPROM-Access - Wrong address range amount of bytes not correct

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7.12.2 Write EEPROM Data

Supported in Mode: 0x85

Tester - Request

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x07	Amount of valid bytes following
5	Service Name	0x3D	Service ID: Read Memory by Address
6	Databyte1	0x0?	Address High Byte
7	Databyte2	0x??	Address Mid. Byte
8	Databyte3	0x??	Address Low Byte*
9	Databyte4	0x03	EEPROM Access
10	Databyte5	0x01	Number of Bytes to be written (1Byte max)
11	Databyte6	0x??	Value to be written
12	Checksum	CS	

ECU – Response if conditions are correct:

Position	Contents	Value	Description
1-3	Header		
4	Length Byte	0x05	Amount of valid bytes following
5	Service Name	0x7D	Service Response ID: Read Memory by Address
6	Databyte1	0x0?	Address High Byte
7	Databyte2	0x??	Address Mid. Byte
8	Databyte3	0x??	Address Low Byte*
9	Databyte4	0x03	EEPROM Access
10	Checksum	CS	

Address-range 0x000 – 0x3FF to be read out via command
"Get Parameter Block Addresses" (chapter 7.9.4)

Negative Response

Current ECU-State / Mode	Valid ECU-RC	Reason
Operational Mode (0x81)	$21 $31	- ECU is executing a control cycle - ECU is executing an EEPROM-Access - Wrong address range amount of bytes not correct
Diagnostic Mode (0x87)	$21 $22 $31	- ECU is executing an EEPROM-Access Conditions not correct (vehicle is not in standstill) - Wrong address range amount of bytes not correct

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